Infinite Series Solutions

Frobenius Method

The method of Frobenius is a way to find an infinite series solution for a second-order ordinary differential equation near a regular singular point.

The following second order differential equation

$$P(x)y'' + Q(x)y' + R(x)y = 0 \qquad \textbf{(1)}$$

has a regular singular point at x_0, then there exists at least one solution of the form

$$y = \sum_{r=0}^{\infty} a_r x^{r+c}$$

$$y = a_0 x^c + a_1 x^{c+1} + a_2 x^{c+2} + \ldots$$

An infinite series of this form is called a Frobenius series.

Following the method is a series of steps.

(i) We seek a solution of the form $y=\sum_{r=0}^{\infty} a_r x^{r+c}$ **.**

(ii)Compute its derivatives

$$\frac{dy}{dx}=\sum_{r=0}^{\infty}(r+c)a_r x^{r+c-1}$$

$$\frac{d^2y}{dx^2}=\sum_{r=0}^{\infty}a_r(r+c)(r+c-1)x^{r+c-2}$$

Substitute the expressions for $y,\ \frac{dy}{dx},\ \frac{d^2y}{dx^2}$ **into equation** (1)**.**

A solution of this form is called a Frobenius-type solution.

(iii) We obtain a series which must be zero.

Equate to zero the coefficient of the lowest power term in x. Since $a_0 \neq 0$, obtain the roots given by quadratic equation known as indicial equation.

Indicial equation is a quadratic polynomial in r.

(iv) If the indicial equation has two real roots r_1 and r_2 such that the difference of two roots is not an integer, then we obtain two linearly independent solutions.

(v) Equate to zero the coefficient of the other powers of x find the values $a_1, a_2, \ldots$ in terms of a_0.

(vi) Compute the solutions for r_1 and r_2.

Example :

$$(2x + x^3)\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6xy = 0 \qquad (1)$$

$$\frac{d^2y}{dx^2} - \frac{1}{(2x + x^3)}\frac{dy}{dx} - \frac{6x}{(2x + x^3)}y = 0$$

$$P(x) = -\frac{1}{(2x + x^3)} \qquad Q(x) = -\frac{6x}{(2x + x^3)} = -\frac{6}{(2 + x^2)}$$

$$xP(x) = -\frac{x}{(2x + x^3)} = -\frac{1}{(2 + x^2)}$$

$$x^2 Q(x) = -\frac{6x^2}{(2 + x^2)}$$

$P(x)$ **and** $Q(x)$ **are finite at** $x = 0$.

$x = 0$ **is a regular singular point.**

Solution

$$y = \sum_{r=0}^{\infty} a_r x^{r+c}$$

Compute its derivatives

$$\frac{dy}{dx} = \sum_{r=0}^{\infty} (r+c) a_r x^{r+c-1}$$

$$\frac{d^2y}{dx^2} = \sum_{r=0}^{\infty} a_r (r+c)(r+c-1) x^{r+c-2}$$

Substituting these expressions to original equation we obtain

$$\sum_{r=0}^{\infty} a_r \left[2(r+c)(r+c-1)-(r+c)\right]\underline{x^{r+c-1}} +$$

$$\sum_{r=0}^{\infty} a_r \left[(r+c)(r+c-1)-6)\right]\underline{x^{r+c+1}} = 0$$

Equating to zero the coefficient of the lowest power of x **when** $r=0$ **i.e.** x^{c-1}**.**

$$a_0\left[c(2c-3)\right]=0$$ **This equation is called the indicial equation.**

$a_0 \neq 0$ $\qquad c = 0, \dfrac{3}{2}$

Equating to zero the coefficient of the next power of x i.e (x^c)

$$a_1\left[(2c-1)(c+1)\right]=0 \longrightarrow a_1=0$$

$$a_2=\frac{-(c^2-c-6)}{(2+c)(1+2c)}a_0$$

The general term

$$a_{r+1}(r+c+1)\left[2(r+c)-1\right]+ \\ a_{r-1}\left[(r+c-1)(r+c-2)-6\right]=0$$

$$a_{r+1}=\frac{\left[6-(r+c-1)(r+c-2)\right]}{(r+c+1)(2r+2c-1)}a_{r-1} \quad r>1$$

$$y = x^c\left(a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots\right)$$

$$a_1 = a_3 \ldots = 0$$

$$y = a_0 x^c\left(1 + \frac{\left(6 - c(c-1)\right)}{(c+2)(2c+1)}x^2 + \ldots\right)$$

$$a_0 \neq 0$$

When c = 0

$$y_1 = A\left(1 + 3x^2 + \ldots\right)$$

When c = 3/2

$$y_2 = Bx^{3/2}\left(1 + \frac{3}{8}x^2 + \ldots\right)$$

General Solution $y = y_1 + y_2$